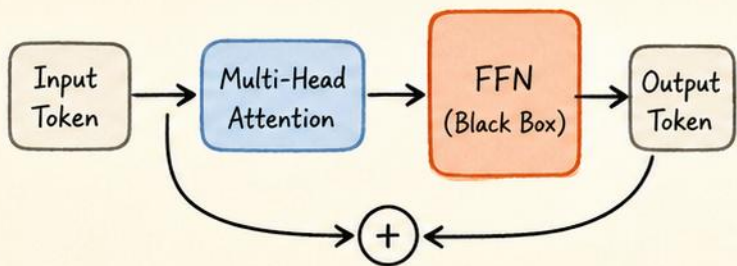


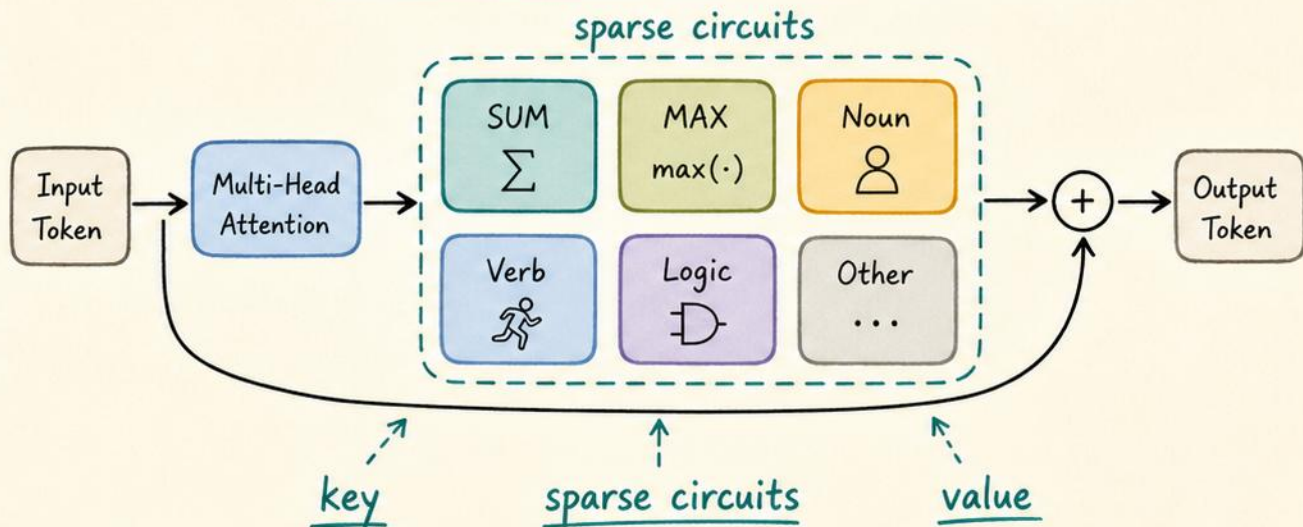
FFN 不是黑盒

The FFN as a Key-Value Memory 解读

传统视角：黑盒 FFN



我们的视角：FFN 由专业电路组成 (可解释、可干预)



稀疏 · 分工 · 可干预

☆ 一个任务：指令决定程序

operand

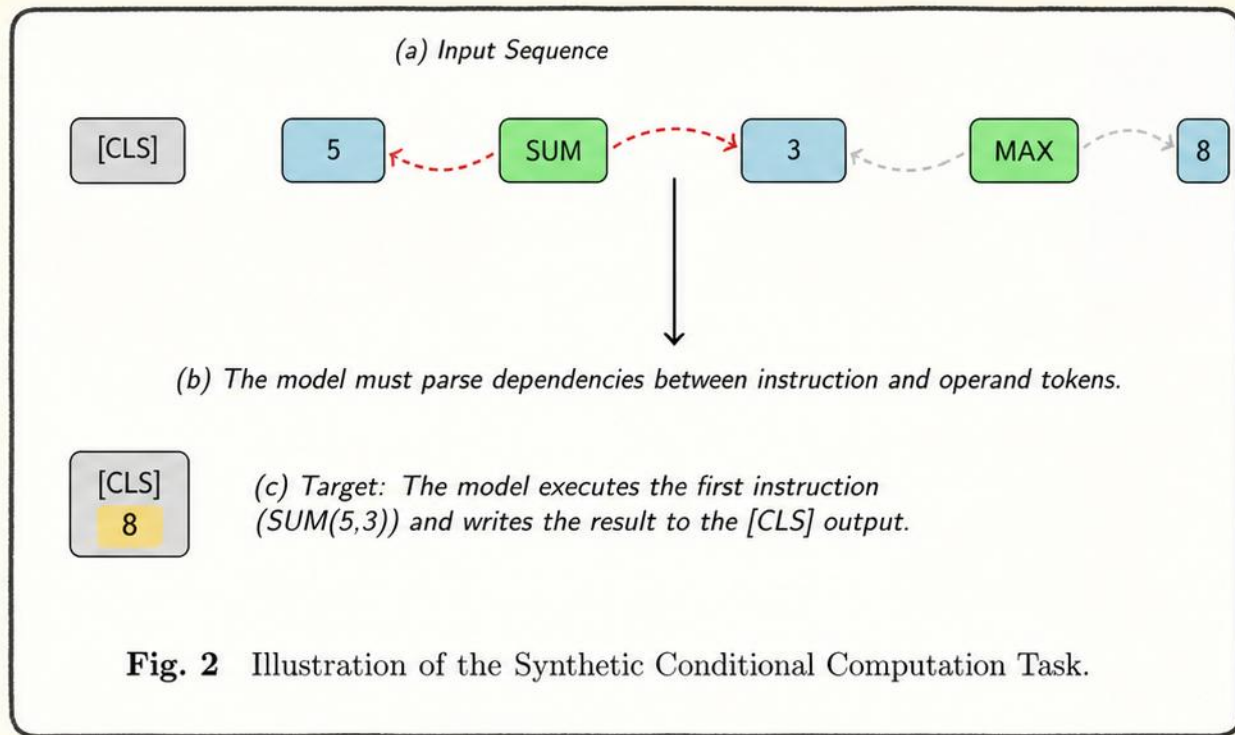
操作数：提供具体的数据。

operator

运算符：决定执行什么操作。

target

目标：指令决定写入 [CLS] 的结果。



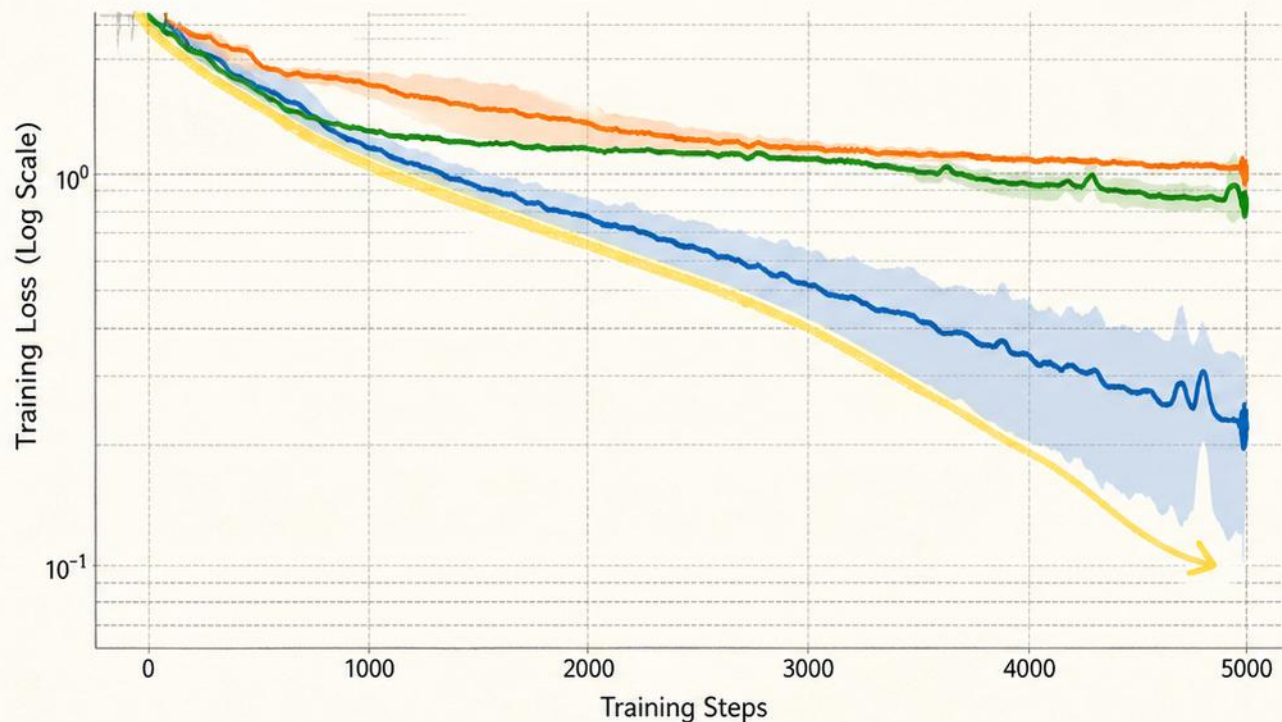
← [CLS] 写答案

模型把计算结果写回 [CLS]。

SUM → 看见 SUM → 求和 MAX → 看见 MAX → 取最大值

只有非线性 FFN 解得动 ☆

Fig. 3 Training loss curves for the Conditional Computation Task.



Each curve shows the mean over 3 random seeds, with shaded regions indicating ± 1 standard deviation. The Standard Transformer (blue) reliably converges, while Attention-Only (orange) and Linear-FFN (green) fail to solve the task.

● Standard 收敛

attention 会看见
FFN 才会计算

● Attention-Only 停滞

attention 会看见
但没有非线性, 无法计算

● Linear-FFN 停滞

有线性映射
缺乏非线性, 无法计算



GELU 是关键
不是参数更多, 而是非线性

FFN 的默认状态是沉默

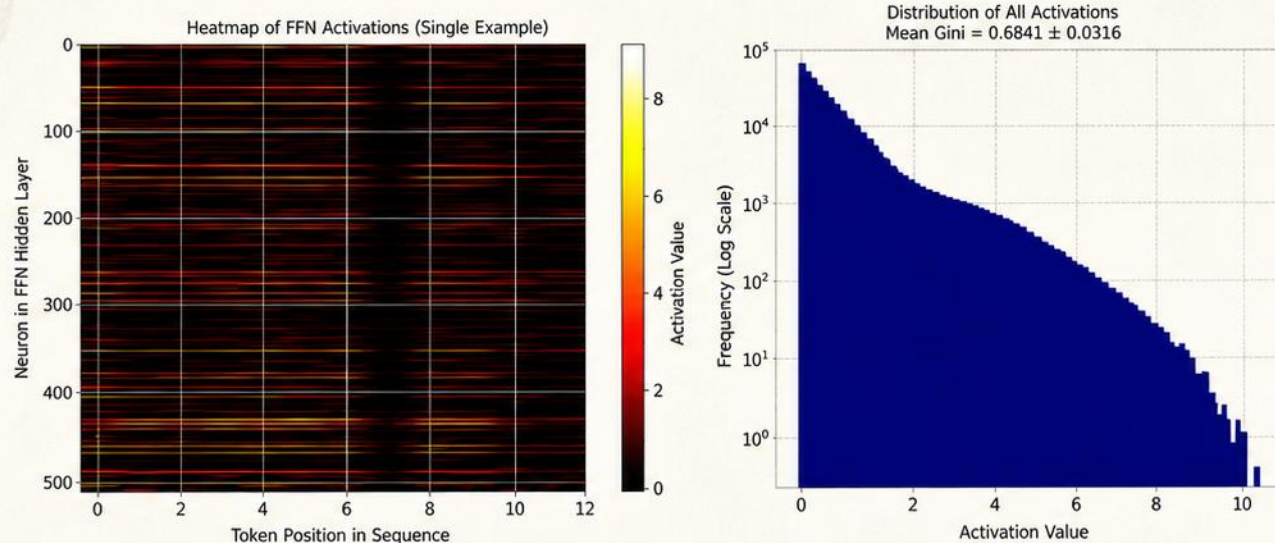
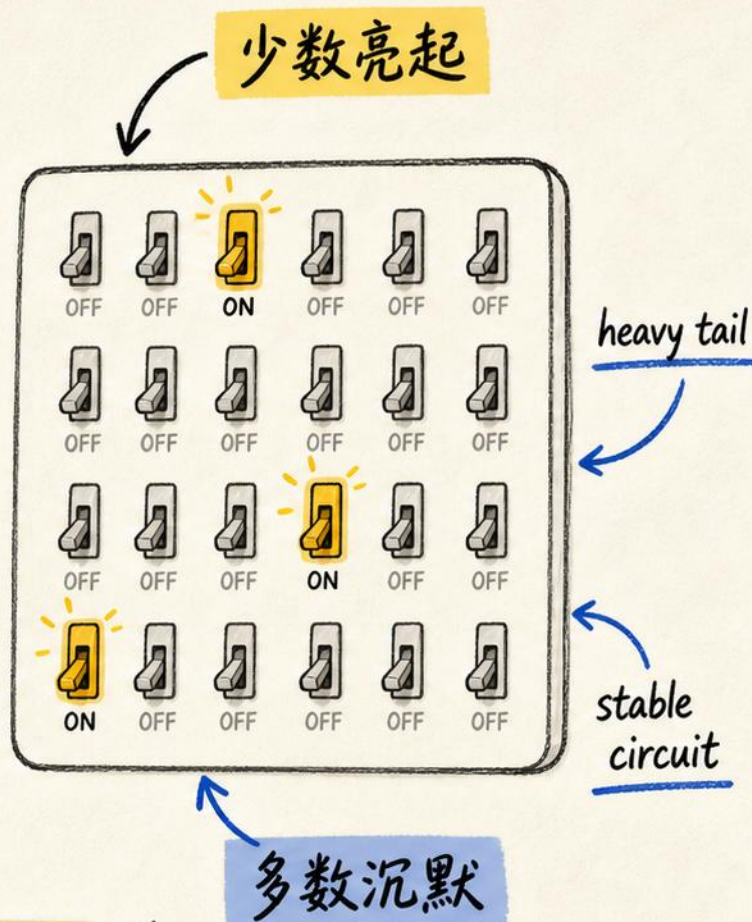


Fig. 4 Internal activations of the FFN. Left: Heatmap of a single representative sequence, showing that only a small subset of the 512 hidden neurons are active across the 12 token positions. Right: Histogram of all non-zero activations on a probe batch (log scale) across three different random seeds, the FFN's Gini coefficient averaged 0.6841 ± 0.0316 , providing robust quantitative confirmation of sparse, heavy-tailed activation.

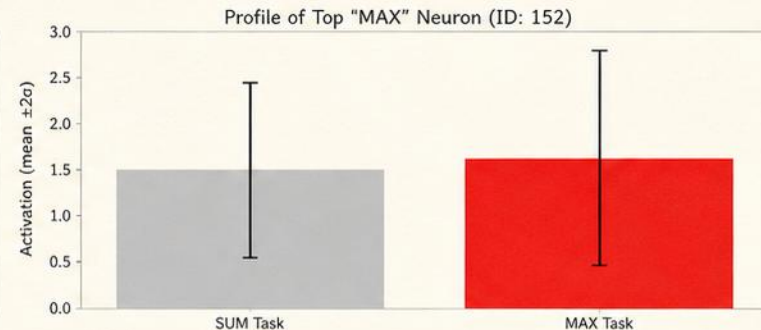
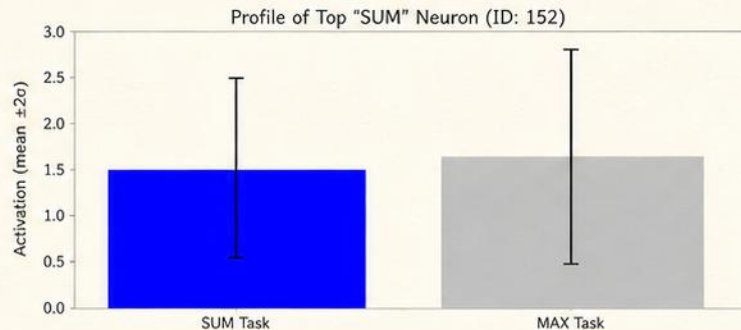
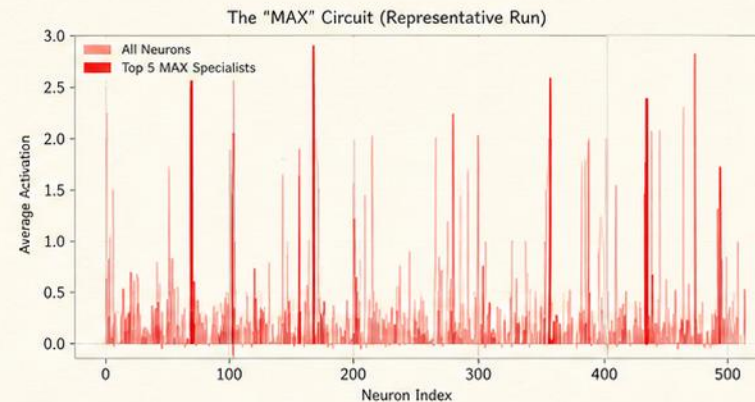
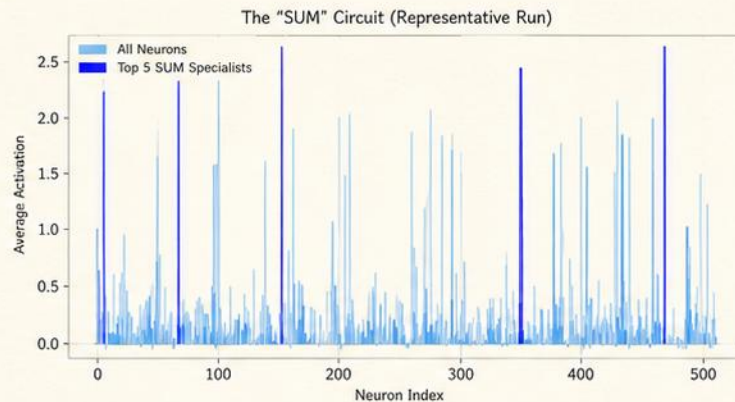


$$\text{Gini} = 0.6841 \pm 0.0316$$

SUM / MAX 有不同 circuit

specialists 不重叠

层级分工



operator key



generalist
router

neuron 152

SUM specialists



MAX specialists



消融给出因果证据

→ double dissociation

	Accuracy on SUM Tasks (%)	Accuracy on MAX Tasks (%)
Unablated Model (Full)	99.6%	99.4%
GR Ablation (None)	10.5%	11.2%
SUM Execution (SEC)	12.1%	98.9%
MAX Execution (MEC)	99.2%	15.3%

GR off →
SUM/MAX 都崩

SEC off →
SUM 崩 / MAX 保持

MEC off →
MAX 崩 / SUM 保持

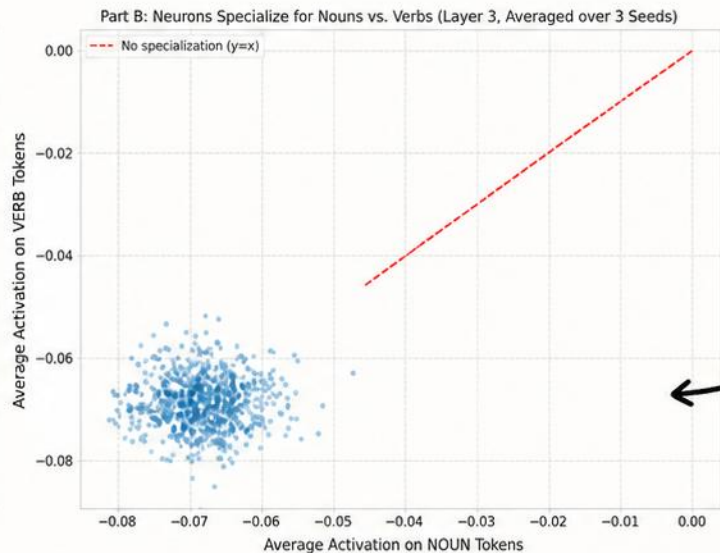
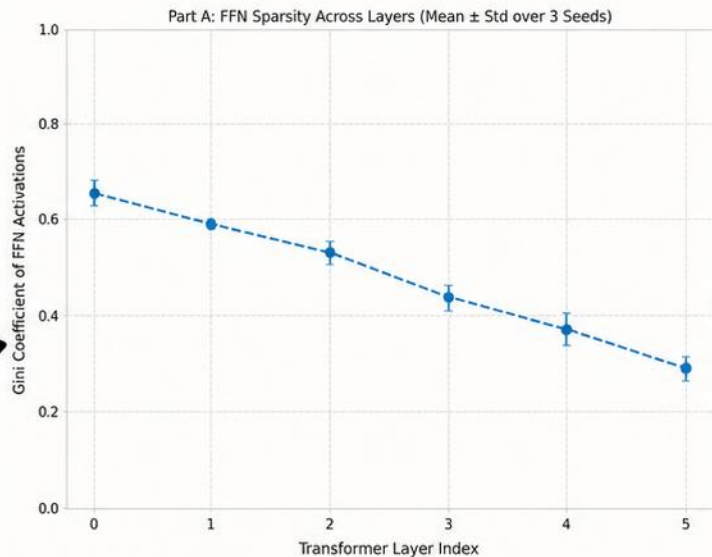
baseline ~99%

fail ~10-15%

DistilBERT 验证

DistilBERT 里也有这件事

浅层更稀疏



Noun / Verb specialists

Layer 3

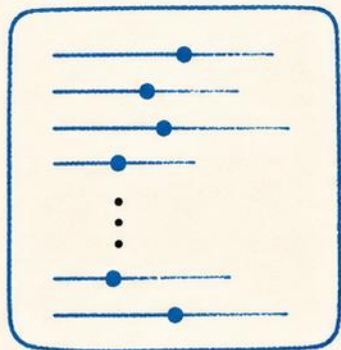
Fig. 6 Validation in a Real-World Model (DistilBERT, Averaged over 3 Seeds). This figure confirms that our key findings on sparsity and specialization extend to a pre-trained language model. All results are averaged over three independent analysis runs with different data samples.



合成任务里的机制，在语言模型里也能看到。

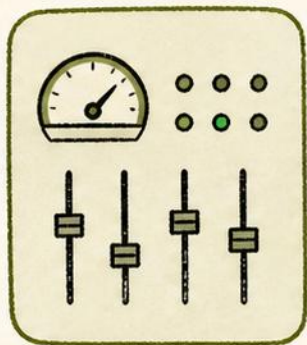
FFN = 可查询的程序记忆

上下文表征 / key



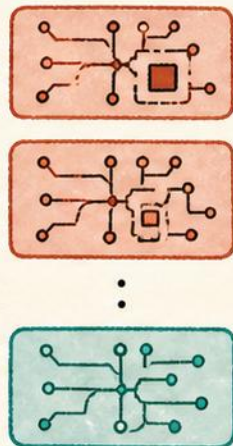
key: 上下文信号

路由器 (通用)

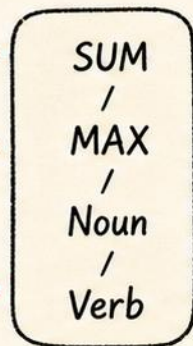


router

专家电路 (专用)



输出 / value



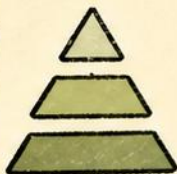
value: 被激活的程序



稀疏



每个上下文只激活少数专家电路。



层级

路由在层间组织，形成层级的专长。



可干预

消融特定专家会改变模型行为。